



A person playing football with C-leg (Courtesy: www.ottobock.com)

grees of freedom has been developed recently, which offers movement beyond the limits of a biological arm. It is equipped with an array of sensors that measure temperature and vibration. The MPL developed by John Hopkins University Applied Physics Laboratory has been successfully tested by University of Pittsburgh School of Medicine and UPMC's brain-computer interface on a neck-down paralysed woman. After implantation, the woman was able to control the MPL with her thoughts alone.

Bionic hand. A bionic hand offers five independently operating fingers, a rotating thumb and a rotating wrist. This lets the patient perform various functions like grabbing a cup, turning a lock and pressing buttons. A bionic hand is equipped with powerful electrical sensors that can detect muscular movements in the residual limb. The activity corresponding to each set of signals is selected from a pre-programmed list. Interestingly, the latest version of a bionic hand, named i-Limb Ultra, developed by Touch Bionics, helps in controlling actions performed by the hand via a smartphone app.

Smart leg. Smart legs are prosthetic legs that offer increased functionality and comfort with the help of a digital control system. According to Dave Kozlowski of Sandia National Laboratories, "The smart legs will enable a more normal and efficient walking gait." Scientists of Sandia are creating a socket that will self-adjust to the swelling of the residual limb over the course of the day. This can avoid the pressure sores that may result from rubbing against the

prosthetic.

C-leg by Otto Bock Healthcare is a prosthetic device with built-in micro-processor that increases the stability and lets amputees tackle slopes, ride bikes and participate in sports activities.

Ankle-foot system. The latest models of ankle-foot system available today use complex-motion sensors and ankle-control software. They act similar to a natural foot and help the amputee to sit, stand and use the stairs and slopes with much more ease than the traditional prosthetic feet.

Internal organ implants

There are a handful of devices that are implanted in/on internal organs like heart, lungs and bladder. They help the organs work normally.

Implantable defibrillator. An implantable cardioverter-defibrillator (ICD) is a small, automatic impulse generator and bi-ventricular pacifier implanted in patients suffering from the risk of ventricular fibrillation, ventricular tachycardia, bradycardia and congestive heart failure. This de-

vice is battery powered, automatically detects cardiac arrhythmia (irregular heartbeat) and creates a jolt of electricity that forces the heart to restart itself with normal rhythm. This implant is widely used by cardiac patients, and the leading manufacturers of this device are Medtronic, Biotronik, Sorin Group CRM, Boston Scientific and St. Jude Medical, Inc.

Deep brain stimulation. Deep brain stimulation (DBS) is a surgical procedure in which the implanted device creates pulses that directly interfere with the neural activity at the targeted site, resulting in the control of functions like movement and mood. This method is an approved treatment for essential tremors, Parkinson's disease and dystonia, and also used to treat epilepsy, chronic pain and depression. Currently, one of the major manufacturers of DBS is Biontronic.

Vagus nerve stimulation. Vagus nerve stimulation (VNS) is a major treatment technique for epilepsy and treatment-resistant depression. Via this therapy, electric pulses are sent to the vagus nerve, the tenth cranial nerve, arising from the medulla region of the brain. Since the vagus nerve is associated with several functions of the brain, researches are being conducted to determine the effectiveness of this method for the treatment of medical conditions like migraines, obesity, fibromyalgia, Alzheimer's, tinnitus and anxiety disorders. Cyberonics, MetaCure and EnteroMedics are the major producers of this implant.

Prosthetics have become much more advanced, sophisticated and user friendly than before. The world portrayed in sci-fi novels like *Cyborg*, films like *Star-Trek* and *Terminator* and TV series like *The Bionic Woman*, may not be far away. The line of separation between humans and machines has thinned incredibly, giving us bright promises of 'an era of technological singularity.' ●

The author is a technical correspondent at EYF Bengaluru